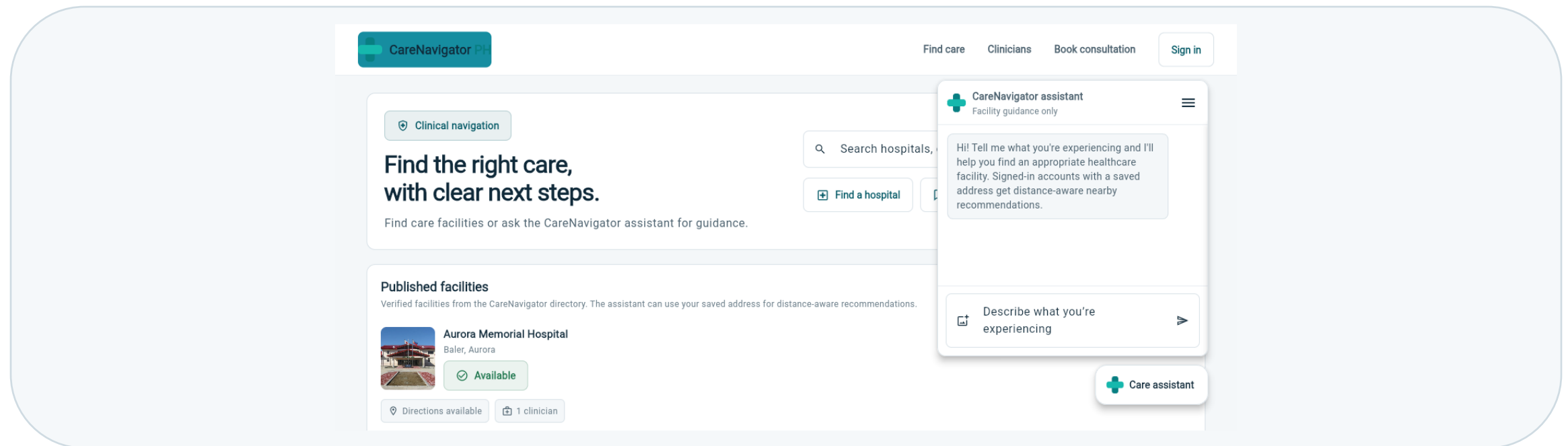




CareNavigator PH

A role-aware healthcare navigation and digital care platform

SYSTEM DEFENSE PRESENTATION



Care access breaks across disconnected steps

Finding a hospital is only the first decision.

- Public hospital, service, clinician, schedule, and capacity information are difficult to compare in one place.
- Guest intake, booking, consultation, and follow-up often require repeated handoffs and duplicate information.
- Clinical records, documents, prescriptions, laboratory results, and reminders can become fragmented across actors.

DESIGN RESPONSE

01 Guide

Verified discovery plus safety-aware assistance

02 Connect

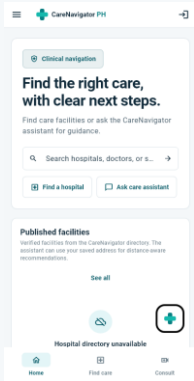
A continuous path from request to consultation

03 Protect

Role-scoped data, private files, and auditability

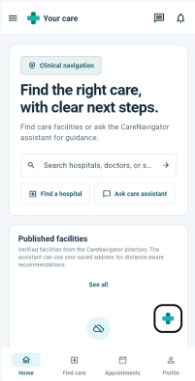
One platform supports five role-aware experiences

The same care journey changes by responsibility; routes, actions, and data visibility follow the signed-in role.



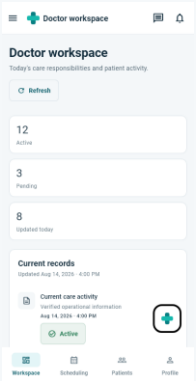
Guest

Discover care, ask the assistant, and submit a verified request



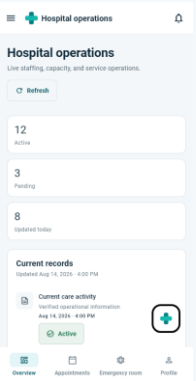
Patient

Book care, communicate, and manage records



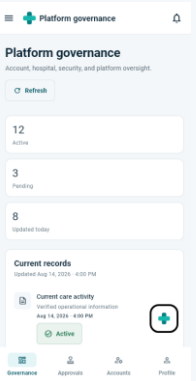
Doctor

Manage schedules, consultations, prescriptions, and labs



Hospital admin

Operate one hospital, staff, services, and capacity



Super admin

Govern hospitals, accounts, settings, analytics, and audits

Six objectives make care accountable

GENERAL OBJECTIVE

Design and implement a secure, responsive platform that helps people find appropriate care and supports the care lifecycle across authorized users.

1 Make care discoverable

Publish verified hospitals, services, clinicians, schedules, routes, and availability.

2 Guide safely

Use deterministic emergency checks and preliminary AI guidance without replacing diagnosis.

3 Connect the journey

Link guest intake, booking, consultation, communication, documents, and follow-up.

4 Enable role-based work

Provide patient, doctor, hospital-administrator, and platform-governance workspaces.

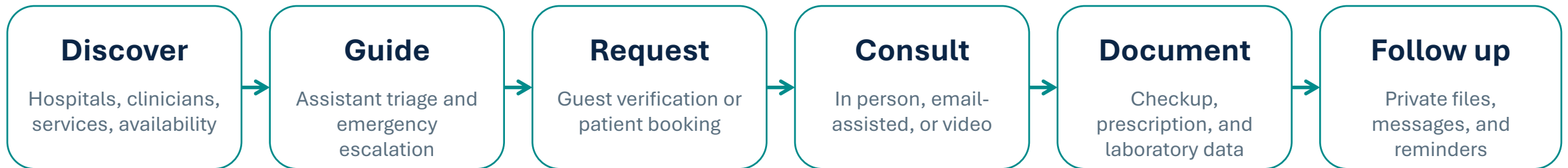
5 Protect clinical data

Enforce authorization with JWTs, RLS, private storage, constraints, triggers, and audit logs.

6 Verify quality

Use analysis, unit/widget tests, SQL acceptance tests, build checks, and visual baselines.

The workflow preserves continuity from discovery to follow-up



Authorization and safety remain continuous

Role guards in the client; RLS, constraints, and server-side checks in the backend; preliminary AI always separated from official clinical decisions.

Scope is broad; production boundaries are explicit

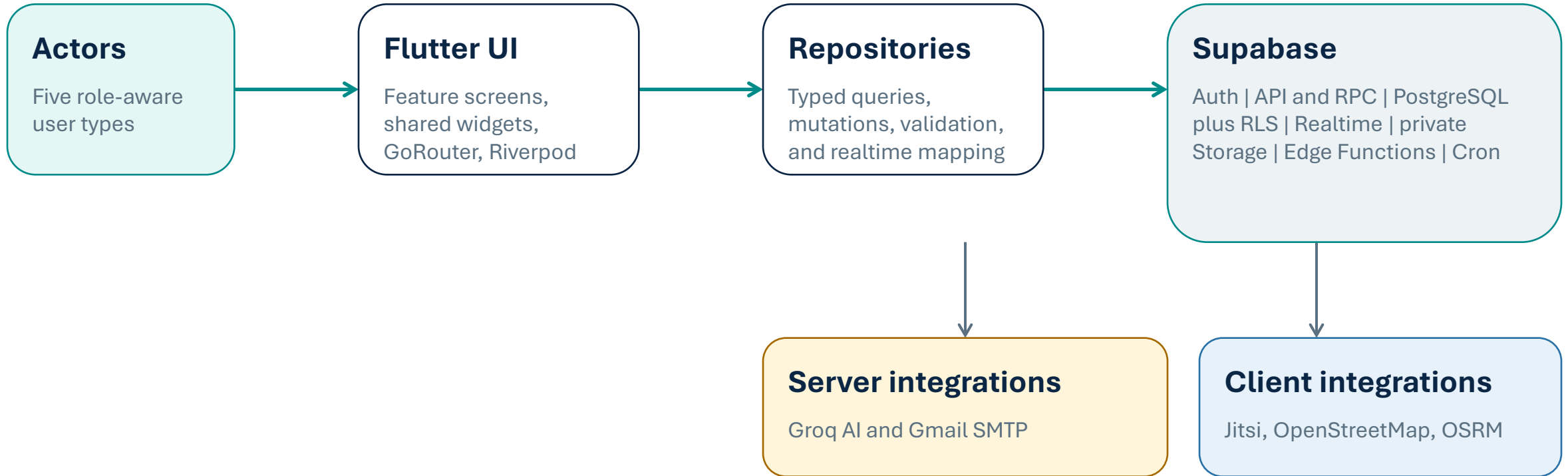
IN SCOPE

- Responsive Flutter mobile and web experiences
- Public hospital and clinician discovery with maps and routes
- Guest intake, patient booking, consultation, messaging, and notifications
- Doctor schedules, structured clinical documentation, prescriptions, and laboratories
- Hospital operations plus platform governance
- Supabase Auth, RLS, Realtime, private Storage, RPCs, Edge Functions, and audit controls

CURRENT LIMITATIONS

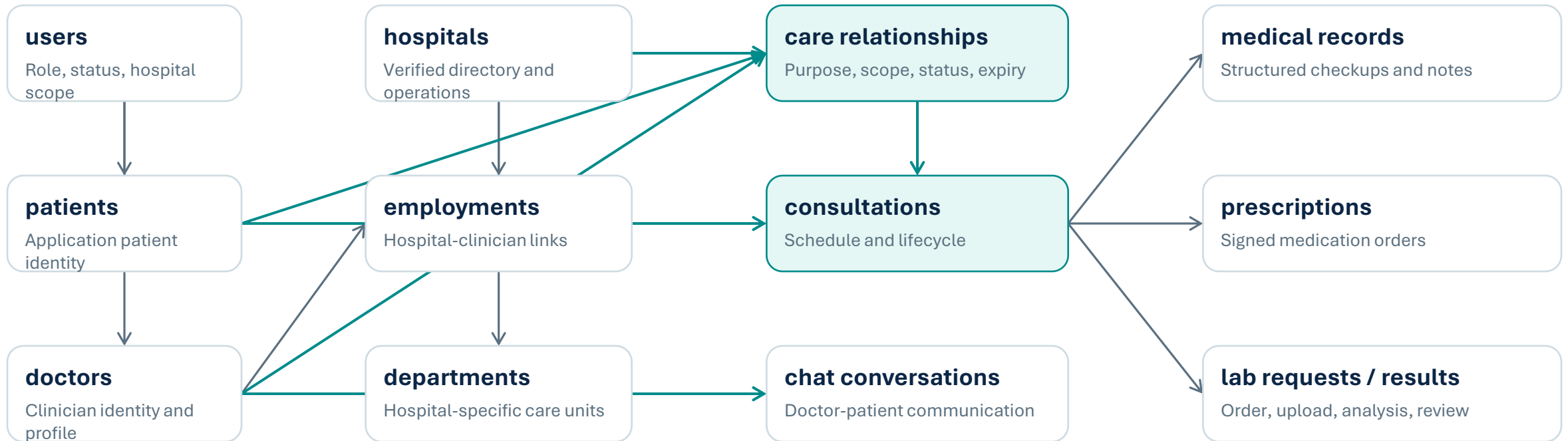
- Care-assistant output is preliminary guidance, not a diagnosis or prescription
- External-provider behavior depends on configured Groq, SMTP, Jitsi, map, and Supabase services
- Live role-to-role end-to-end tests require seeded non-production accounts and credentials
- iOS/device, load, disaster-recovery, privacy, and compliance validation remain before release
- Public directory screenshots show explicit unavailable states when backend configuration is absent
- Current visual regression baselines require review after recent UI changes

Layers separate presentation, data access, and authority



Public configuration stops at the client. Privileged keys and clinical authorization remain server-side.

Care relationships govern clinical access



Full field-level Mermaid ERD: [docs/defense/database_erd.mmd](#)

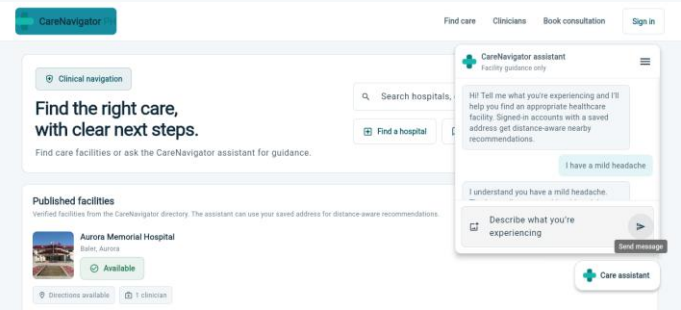
RLS policies, grants, triggers, functions, indexes, audit logs, and private Storage policies complete the physical design.

Security is enforced below the interface

The UI improves usability. It is not the authorization boundary.

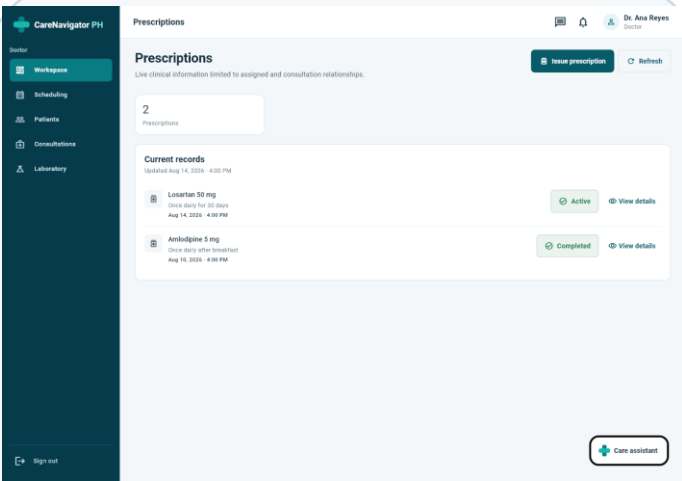
- 1 Role-aware routes and guarded actions** Presentation layer prevents invalid navigation and exposes clear account states
- 2 Typed repository boundary** Validation, exact IDs, controlled mutations, and explicit failure handling
- 3 JWT plus PostgreSQL Row Level Security** Patient, doctor, hospital, relationship, and action scope are checked by the backend
- 4 Private Storage and signed URLs** Clinical and identity files are never permanent public assets
- 5 Constraints, triggers, audits, and server-only secrets** Clinical invariants and privileged operations remain observable and accountable

Implemented screens span care and operations



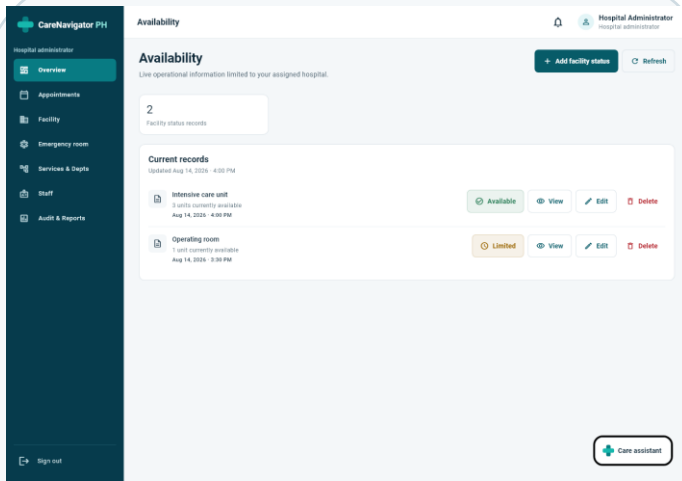
Care assistant

Safety-aware guidance with facility context



Doctor prescription workflow

Reviewable structured medication orders



Hospital operations

Connected availability and capacity controls

Current build evidence is strong - but not fully green

Verified locally on 02 September 2026

0

analyzer issues

243

tests passed

11

live tests skipped

63

migration files

Visual regression status

8 passed | 8 failed

The failures are golden-image differences after recent UI changes. They require intentional review and baseline refresh - not concealment - before claiming a fully green suite.

Progress moved from foundation to integrated care

09 AUG

Foundation verified

Flutter architecture, role routing, live Supabase contract, public discovery, auth, and initial workspaces.

23-26 AUG

Care workflow expansion

Multi-hospital authorization, online requests, patient linking, structured prescriptions and diagnostics, persistent messaging, reminders.

02 SEP

Current defense state

Care assistant integration, 63 migrations, clean static analysis, broad automated coverage, and identified golden-test drift.

Current state: a functioning integrated prototype with verified strengths and explicit gaps.

Four actions remain before production

- | | | |
|-----------|--|---|
| 01 | Review visual drift | Confirm intended UI changes, then update only approved golden baselines. |
| 02 | Run live role journeys | Use seeded non-production accounts for patient-to-doctor-to-admin end-to-end tests. |
| 03 | Validate providers and devices | Exercise Groq, SMTP, Jitsi, scheduled jobs, Android hardware, and iOS/macOS builds. |
| 04 | Formalize operations and compliance | Complete privacy/security review, monitoring, load tests, backups, retention, recovery, and key rotation. |

CareNavigator PH demonstrates a viable, secure path from finding care to managing care.

The prototype is defensible because its value, architecture, authorization boundaries, database relationships, implemented workflows, and remaining risks are all visible and testable.

Panel decision requested

Approve continued development toward live end-to-end validation and production-readiness controls.